Robust Reward-Free Exploration under Distributional Drift

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***Abstract*—Reward-free exploration ("RFE") decouples exploration from reward specification, enabling agents to learn environmental representations and contextual information that can later be reused for downstream tasks. While existing RFE methods provide strong theoretical guarantees under stationarity assumptions, their robustness in environments that evolve over time remains underexplored. In this paper, we evaluate reward-free exploration in non-stationary Markov decision processes ("MDPs") by simulating environmental drift, including reward shifts, transition noise, and structural changes to the environment. Using a controlled and tunable GridWorld testbed, we compare the performance of fixed and drift-aware state representation encoders during the exploration phase. Our results indicate that representation choice plays a critical role in robustness: fixed representations often perform well when drift affects rewards alone, whereas drift-aware representations exhibit slight improved performance when transitions or environment structure change. Motivated by these findings, we propose that the RFE framework should adapt representation learning based on the observed type of drift, improving stability and return under non-stationary conditions.**

***Index Terms*—Domain Shift, Reward-Free Exploration, State Encoder, Markov Decision Process**

I. Introduction

Reinforcement learning ("RL") has demonstrated remarkable success in domains ranging from game playing to robotics and decision-making systems. However, a persistent challenge in RL is efficient exploration, particularly in environments where reward signals are sparse, delayed, or unavailable during the data collection phase. Reward-free exploration ("RFE") addresses this challenge by explicitly decoupling exploration from reward specification, allowing agents to collect transition data and learn reusable representations of the environment before any downstream task or reward function is defined. This paradigm offers strong theoretical guarantees for sample efficiency and flexibility, making it a promising framework for general-purpose RL.

Most existing reward-free exploration methods are developed under the assumption that the environment is held static, i.e., the underlying transition dynamics and environment structure are assumed to remain fixed throughout both the exploration and planning phases, and often continuing even further in time; however, in real-world settings, many environments evolve over time, and this assumption reduces RFE's theoretical applicability. In an effort to understand how reward-free exploration behaves under distributional drift, it is critical to deploy these methods in realistic settings and observe the outcomes.

In this work, we aim to study RFE in non-stationary Markov Decision Processes by introducing controlled forms of environmental drift in a tunable environment. In the scope fo this paper, we considered three types of drift that commonly arise in practice: reward shifts, transition noise, and structural changes to the environment. These drift types affect different components of the MDP and allow us to isolate how various aspects of drift interact with reward-free exploration. Our study is conducted using a GridWorld environment control over drift magnitude and timing.

A central focus of this paper is the role of state representation, done through two types of state-encoders. We compare two exploration strategies that differ in how representations are learned during the exploration phase: a fixed encoder that remains unchanged across time, and a drift-aware encoder that continues to adapt as the environment evolves. While fixed representations can offer stability and reduced variance, adaptive representations may better capture changes in transition dynamics or environment structure. Through empirical evaluation, we show that representation choice may impact robustness under drift. Fixed representations perform well when drift affects rewards alone, whereas drift-aware representations provide improved performance when transitions or structural properties of the environment change.

Contributions. The main contributions of this paper are:

- Evaluating of reward-free exploration under multiple forms of environmental drift.
- A comparison of fixed and drift-aware state representation strategies during the exploration phase.
- Observations that suggest RFE framework may improve robustness and performance curating its approach to the type of drift.

II. Related Work

A. Reward-Free Exploration

Reward-free exploration was formalized as a reinforcement framework by by Jin et al. [1] In this paradigm, an agent first performs an exploration phase without access to any reward signal, with the goal of collecting transition data sufficient to construct an empirical model of the environment. One exploration is complete, an arbitrary reward function

is specified, and planning is performed using the previously collected data of the environment. This decoupling enables strong theoretical guarantees on sample efficiency and allows exploration data to be reused on multiple downstream tasks. Furthermore, it's been shown that RFE can achieve near-optimal sample complexity for episodic MDP through structured exploration policies that maximize coverage of the state-action space. [1] However, this prior analysis relies on an assumption of stationarity, that is, the transition dynamics and environment structure are assumed to remain fixed throughout both exploration and planning. While this assumption is standard in RL theory, it severely limits the applicabilty of RFE methods in environments that evolve over time. Our work builds directly on this framework by studying how violations of stationarity (through introducing controlled drift) effect the performance of RFE. Recent work has also highlighted the importance of representation learning in reinforcement learning, particularly in settings where rewards are sparse or delayed. Studies [4] demonstrate that learning effective state representations thorugh contrastive objectives can significantly improve sample efficiency and downstream performance. While their work is not situated in the reward-free setting, it motivated our focus on representation learning during the exploration phase. In this paper, we examine how different representation strategies, fixed vs. drift-aware encoders, interact with reward-free exploration under drift.

B. Distributional Domain Shift

Robustness against distributional shift has been extensively studied in both optimization and ML. Besbes et al. [3] analyze stochastic optimization problems in non-stationary environments, providing regret bounds under drifting distributions. Their work formalizes the impact of temporal changes in the underlying data-generating process and underscores the challenges posed by non-stationarity. Although their analysis is not specific to reinforcement learning, it provides a theoretical foundation for understanding performance degradation under environmental drift. In supervised learning, robustness to distributional change has been studied through the lens of domain adaptation. Xu et al. [2] investigate robust multi-source domain adaptation under label shift, proposing methods that explicitly account for changes in the output distribution. This work emphasizes that different sources of distributional shift require distinct adaptation strategies, an insight that informs our treatment of environmental drift in reinforcement learning. In the context of reinforcement learning, distributional drift commonly manifests as non-stationarity in rewards, transition dynamics, or environment structure. Prior studies have shown that RL agents often fail to generalize when evaluated under shifted conditions [5]. However, most existing approaches to non-stationary RL tightly couple exploration with reward optimization, making them difficult to reconcile with reward-free paradigms. In contrast, our work isolates non-stationarity during the reward-free exploration phase and systematically studies how different forms of drift interact with representation learning. By drawing connections between robustness under domain shift [2], [3] and representation learning in RL [4], we aim to test whether adapting representations to the source of drift can improve the robustness of reward-free exploration.

III. Methodology

A. Problem Setting

We consider episodic Markov decision processes (MDPs) defined by the tuple $(\mathcal{S}, \mathcal{A}, P_t, r_t, H)$, where $\mathcal{S}$ is the state space, $\mathcal{A}$ is the action space, $P_t(s' \mid s, a)$ denotes the (possibly time-varying) transition dynamics, $r_t(s, a)$ is a reward function specified only after exploration, and H is the episode horizon. In contrast to the stationary setting typically assumed in reward-free exploration, we allow the environment to evolve over time, resulting in non-stationary transition dynamics or reward functions. This is accomplished through a scheduled series of controlled drifts that are specified in the subsections below.

Our goal is to study reward-free exploration under distributional drift. During the exploration phase, the agent does not observe rewards and collects transition data by interacting with the environment. After exploration, a reward function is specified and a policy is computed using the collected data. To evaluate the quality of the learned transition model and representations, we introduce a sparse goal-reaching reward function only after the exploration phase is complete. Specifically, our reward function assigns a unit reward when the agent reaches a designated goal state and zero reward otherwise. Performance is evaluated by the return achieved under the post-exploration policy in the presence of environmental drift.

B. Reward-Free Exploration Framework

We adopt the reward-free exploration framework introduced by Jin et al. [1]. In this framework, exploration and planning are decoupled into two distinct phases. During exploration, the agent executes a policy designed to sufficiently cover the state-action space and collects transition tuples (s, a, s'). In this phase, there is no reward information used, instead our agent employs a simple ϵ-random coverage-seeking exploration policy in this absence of rewards. This ensures persistent stochasticity in action selection and encourages broad visitation of states even as the environment changes. To study violations of stationarity, we do allow the environment to change during exploration by updating the active MDP at scheduled drift times by an index time.

After exploration, we enter the planning phase, where we specify a sparse goal-reaching reward and compute a policy using the learned transition model from the prior phase. We then evaluate the post-exploration policy in environments exhibiting the same drift type to measure how well the exploration data and learned representations support downstream control under drift.

C. GridWorld Environment

To study reward-free exploration under controlled distributional drift, we employ a configurable GridWorld environment that allows explicit manipulation of both transition dynamics

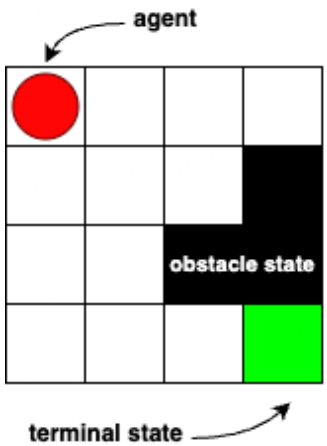

Fig. 1. Visual of a GridWorld environment layout

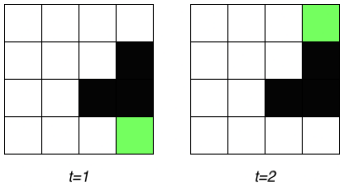

Fig. 2. Visual of Reward-Shift: Moving the terminal states in the evaluation phase.

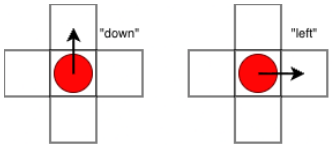

Fig. 3. Visual of Transition-Noise: The intended action has a tunable probability to cause an unintended transition.

and environment structure. The environment is defined as a two-dimensional grid composed of free cells, obstacles, and designated terminal states. The agent occupies a single grid cell at each time step and selects an action from a finite action set $\mathcal{A} = \texttt{up}, \texttt{down}, \texttt{left}, \texttt{right}$. Under the base environment configuration, transitions are fully deterministic: each action moves the agent in the corresponding direction unless blocked by an obstacle or boundary, in which case the agent remains in its current place.

Episodes terminate either upon reaching a terminal (goal) state or after a fixed horizon. To enforce reward-free exploration, terminal states are treated as ordinary free cells during the exploration phase, and no reward signal is exposed to the agent. This allows the agent to traverse the environment without receiving any task-specific feedback while still enabling goal-directed evaluation after exploration.

This deterministic and interpretable setting provides a stable baseline for reward-free exploration, enabling precise isolation of distributional drift effects. By selectively introducing controlled perturbations to rewards, transition dynamics, or environment structure, we evaluate how different exploration representations generalize when the underlying MDP deviates from stationarity. In the subsection below, we precisely describe how environmental drift is simulated within this environment.

D. Modeling Environmental Drift

To evaluate reward-free exploration under drift conditions, we simulate controlled forms of environmental drift that modify specific components of the underlying MDP over time. Drift is introduced according to a predefined schedule during the exploration phase, resulting in transition data collected from multiple environment regimes rather than a single stationary MDP. We consider three distinct types of drift, each affecting a different aspect of the environment.

Reward Shift: Under reward shift, the reward function used for evaluation changes over time. In our GridWorld setting, this corresponds to changing the location of the goal state after exploration. During exploration, no reward signal is available, thus reward shift does not alter the dynamics of this phase.

Transition Noise: Transition noise introduces stochasticity into otherwise deterministic environment dynamics. Specifically, with a tunable strength/probability, the intended action made by the agent may result in an unintended transition to a neighboring state. For instance, there is a variable probability that the action `left` may instead cause a transition to the above, below, or rightward neighboring cell rather than the intended leftward cell. This perturbation models uncertainty or degradation in the system dynamics and directly effects the accuracy of the transition model learned under exploration. Transition noise alters the agent's observed transtion distribution and challenges the assumptions that collected dynamics will remain consistent over time.

Structural Drift: Structural drift modifies the layout of the environment by adding or removing obstacles within the GridWorld. These changes alter the connectivity of the state space and can affect both reachability and transition dynamics. Structural drift represents a more severe form of non-stationarity, as it changes the topology of the environment and invalidates portions of the previously collected transition data. This setting tests the ability of exploration representations to adapt when the underlying structure of the environment evolves.

Each drift type can be applied independently to isolate its effects, or in combination, and with controlled timing and

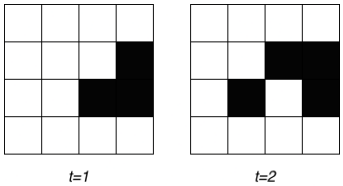

Fig. 4. Visual of Structural-Drift: Topology of the environement changes by adding or removing obstacles.

magnitude.

E. State Representation Learning

A central component of our study is the role of state representation learning during the exploration phase. We compare two representation strategies:

Fixed Encoder: A state encoder trained during an initial exploration period and then held fixed for the remainder of training. While fixed representations can preserve consistency when the underlying dynamics remain unchanged, they may fail to capture shifts in transition dynamics or environment structure introduced by drift.

Drift-Aware Encoder: A state encoder that continues to update throughout exploration as new transition data is collected. This allows the learned representation to adapt to changes in the environment, potentially capturing evolving dynamics or structural modifications. However, continual adaptation may introduce instability or unnecessary variance.

The drift-aware encoder favors adaptive and dynamic representations that respond to changes in environment dynamics or structure, but may introduce instability due to continual updating. In contrast, the fixed encoder favors representational consistency and stability, which can be advantageous but may lead to inaccurate or outdated representations under environmental drift.

IV. Discussion

Our results highlight the importance of representation learning choices in reward-free exploration under non-stationary environments. By evaluating fixed and drift-aware encoders across multiple forms of environmental drift, we observe a clear trade-off between representational stability and adaptability, with performance strongly dependent on the source of drift.

In our primary experiments, we consider a 6×6 GridWorld environment with a single designated goal state. The agent begins each episode from a fixed initial position, and episodes terminate either upon reaching the goal or after a fixed horizon of H steps. During the exploration phase, reward signals are removed by treating the goal state as a free cell, ensuring that exploration remains strictly reward-free. To evaluate robustness under non-stationarity, we introduce environmental drift during the exploration phase. In the reported results, the drift-aware encoder is exposed to a combined drift setting, which includes reward shift, transition noise, and structural drift applied simultaneously at a fixed drift strength.

The observed performance differences reflect a fundamental trade-off between representational stability and adaptability. Fixed encoders produce consistent, low-variance representations that are well suited to stationary environments or reward-only drift. However, when the environment evolves, these representations encode outdated assumptions that degrade downstream planning. Drift-aware encoders, by contrast, continually adapt to new transition data, allowing them to track changes in dynamics or structure at the cost of potential instability when adaptation is unnecessary. The uncertainty of the exploration phase due to the inclusion of scheduled drift makes our drift-aware encoders hold less confident state-encoding, and their performance is partially diminished as a result. However, under structural drift, the performance drop of drift-aware encoders tends to be less severe than strictly fixed encoder.

A. Limitations

This study is limited to discrete GridWorld environments with controlled drift mechanisms. While this setting allows precise isolation of drift effects, it does not capture the full complexity of high-dimensional or continuous domains. Additionally, drift is introduced according to a known schedule; detecting and responding to drift online remains an open problem. Future work could explore principled drift-detection mechanisms, extend these experiments to more complex environments, and investigate theoretical guarantees for reward-free exploration with adaptive representations under non-stationarity.

V. Our Novel Application / Further Proposed Improvements

While reward-free exploration provides strong theoretical guarantees under stationarity, our results demonstrate that its effectiveness in non-stationary environments depends critically on how state representations are learned during exploration. Existing reward-free frameworks typically assume a fixed exploration procedure and do not explicitly account for environmental drift or the type of drift. Our approach is novel because we used two different state encoders as well as an operating RFE paradigm under an evolving environment.

Based on our results, we propose that future work could offer an improvement that takes reward-free exploration with an adaptive representation learning strategy which selectively updates state representations during exploration. Rather than committing to a single representation approach, the agent dynamically adjusts whether representation learning is enabled based on the nature of observed drift. When drift primarily affects rewards, representations are held fixed to preserve stability and reduce variance (operating similar to our fixed encoder style). When drift affects transition dynamics or environment structure, representation learning is re-enabled to allow the encoder to adapt to evolving dynamics (operating similar to our drift-aware encoder style). Continued work with this approach would take the benefits of both stable representations under minimal shift and adaptable representations under aggressive shift.

VI. Results

In this section, we evaluate the performance of the Fixed and Drift-Aware encoders across the three isolated drift types (`GOAL_SHIFT`, `TRANSITION_NOISE`, `WALL_CHANGE`) and a fourth `COMBINED` scenario. We report the average episodic return achieved by the agent during the evaluation phase (post-drift).

A. Performance under Reward Shift

In the GOAL_SHIFT scenario, the agent's transition dynamics and environment topology remained constant, but the location of the terminal reward was moved at Step 3000.

As illustrated in Fig. 5, both encoders successfully adapted to the new goal location; however, the Fixed Encoder demonstrated superior stability. The Fixed Encoder achieved a post-drift average reward of 0.81, compared to 0.80 for the Drift-Aware Encoder. While the performance gap is narrow, the rolling average plot (Fig. 5, left) reveals that the Drift-Aware encoder exhibited higher variance immediately following the drift event. This suggests that when the underlying physics of the environment are stationary, the "plasticity" of the Drift-Aware encoder introduces unnecessary representational noise, whereas the Fixed Encoder benefits from utilizing its stable, pre-drift topological map.

Fig. 5. Performance under GOAL_SHIFT. The Fixed Encoder (red) maintains higher stability and a higher average return (0.80) compared to the Drift-Aware Encoder (0.77), indicating that representational updates are unnecessary when dynamics remain static.

B. Performance under Transition Noise

We subsequently introduced stochasticity into the environment's actuation. Under TRANSITION_NOISE, the intended action resulted in a random transition with probability $p = 0.2$.

Fig. 8 highlights a significant inversion in performance. The Fixed Encoder, relying on a deterministic map learned during the initial exploration, failed to account for the stochastic transitions, causing its performance to collapse to an average return of 0.60. Conversely, the Drift-Aware Encoder rapidly updated its internal model to reflect the increased uncertainty, achieving a robust return of 0.96. This confirms that when the reliability of the agent's actuators degrades, a static representation is insufficient, and continuous learning is required to maintain optimal control.

VII. Conclusion

In this work, we studied reward-free exploration in the presence of distributional drift, with a particular focus on how state representation learning influences robustness under non-stationary environments. Using a controlled and tunable GridWorld setting, we evaluated RFE across multiple simulated drift types, including reward shift, transition noise, and structural changes, and compared the fixed and drift-aware representation strategies during the exploration phase. Our results show that the standard stationarity assumptions

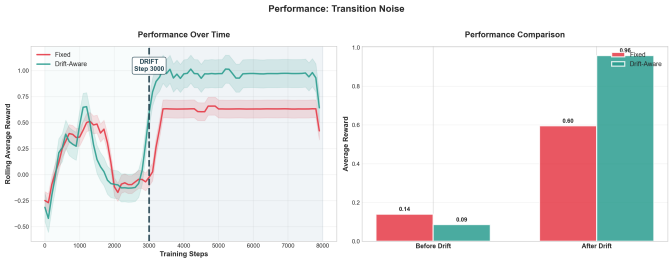

Fig. 6. Performance under TRANSITION_NOISE. The Fixed Encoder's performance degrades significantly (0.6) as it fails to model stochasticity, while the Drift-Aware Encoder adapts to the noise, maintaining high returns (0.96).

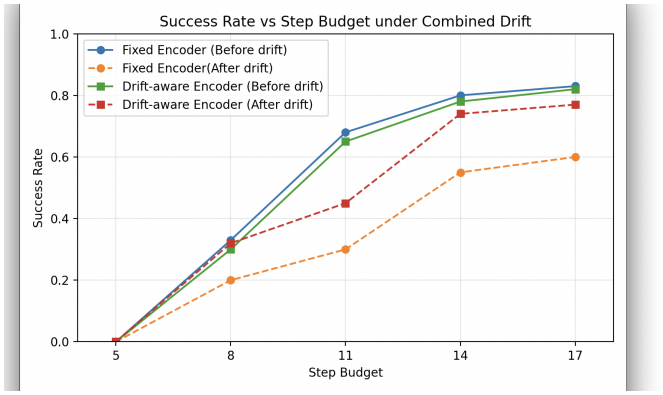

Fig. 7. Success rate of encoders reaching goal with varying step budgets before and after aggressive combined drift averaged across 100 different seeds, 6x6 grid. Drift-aware encoder demonstrates greater performance and stability.

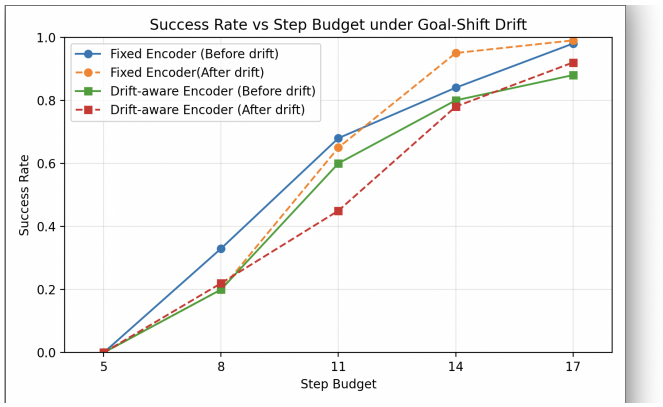

Fig. 8. Success rate of encoders reaching goal with varying step budgets before and after goal-shift drift averaged across 100 different seeds, 6x6 grid. Fixed encoder demonstrates higher performance than drift-aware encoder.

underlying RFE can break down under environmental drift, leading to degrading downstream performance. We find that fixed representations remain effective when drift is limited to rewards, but struggle when broader structural drift occurs, revealing larger performance drops. In contrast, drift-aware representations improve robustness by adapting to changes in the environment, resulting in smaller performance drops after drift at the cost of some representational instability and inconsistent performance gains in less dramatic drift scenarios. Overall, this study suggests that extending RFE beyond stationary assumptions requires careful treatment of representation learning, crucially dependent on the type of drift present to perform optimal state encoding. Future work can explore drift detection mechanisms to dynamically decide which representations best encode the environment for further performance gains.

VIII. External References

[1] C. Jin, Z. Yang, Z. Wang, and M. I. Jordan, "Reward-Free Exploration for Reinforcement Learning," in *Proceedings of the 37th International Conference on Machine Learning (ICML)*, 2020.

[2] Y. Xu *et al.*, "Robust Multi-Source Domain Adaptation under Label Shift," *arXiv preprint*, 2025.

[3] O. Besbes, Y. Gur, and A. Zeevi, "Non-stationary stochastic optimization," *Operations Research*, vol. 63, no. 5, pp. 1227–1244, 2015.

[4] M. Laskin, A. Srinivas, and P. Abbeel, "CURL: Contrastive Unsupervised Representations for Reinforcement Learning," in *Proceedings of the 37th International Conference on Machine Learning (ICML)*, 2020.

[5] C. Packer *et al.*, "Assessing Generalization in Deep Reinforcement Learning," *arXiv preprint arXiv:1810.12282*, 2018.